

A small portion of a tray of MLCCs that have been imaged acoustically is shown in Fig. 2. Echoes were collected simultaneously from all depths within the capacitors to make this image. The result is that all air-gap defects are visible, but there is no information in this image to determine their order to evaluate their depth. So, acoustic cross-sections can be made or a separate image can be made of each step.

Three of the six capacitors have air-type defects. Four of the defects are small voids, but the fifth is a massive delamination incorporating an unknown number of layers within the capacitor. The two capacitors in this image having only small defects may at this point be electrically viable, but the fifth capacitor almost certainly is not.

Large quantities of MLCCs in trays are imaged rapidly by automated AMI tools. MLCCs identified as harbouring defects are removed before a lot of the capacitors is used in assembly.

Fig. 3 shows the acoustic images of two (out of six) levels that were imaged in the same MLCC. To make a separate acoustic image from each depth, a gate was set telling the echo receiver in the transducer to use the echoes whose arrival times indicate that they originated between depth A and depth B to make a separate image.

The echoes that arrive in the first sixth of the total time span are assigned to gate 1 and displayed as a separate image. Those in the next sixth are assigned to gate 2, and so on. The result is an image sequence showing the interior of the capacitor in six evenly thick slices. If needed, there can be hundred or more slices, and they can have different thicknesses, ranging down to extremely thin. When it is necessary to know what features exist at specific depths within a component, this is a very useful imaging mode.

The plates in each of the two stacks in this capacitor are staggered. If the two plates in gate 4 (at the bottom) were pushed together, they would have the same orientation as

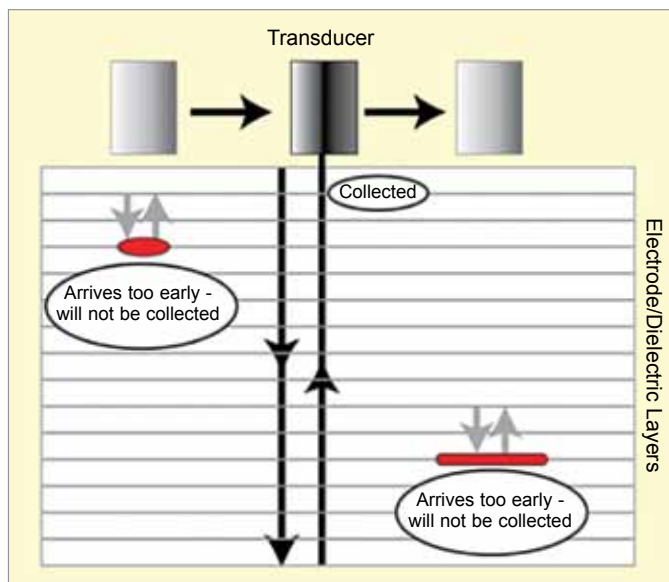


Fig. 4: Imaging of internal defects as shadows

the plates in gate 3 (at top).

Overall, this capacitor has too many voids to be considered for assembly. Most of the voids are seen in only one of the two images, but a few that probably lie close to or at the boundary between the two plates can be seen in both the images.

There are also a few darker features that are the acoustic shadows of voids or delaminations, which lie above a given gate. Two of them are marked by red and green dots. The two marked items must lie above, in gates 1 or 2.

Here is what has happened: a pulse from the transducer strikes a feature in gate 1 or gate 2 and is reflected to the transducer, thereby 'using up' the echo from that x-y position. No ultrasound travels deeper at that position, so when the transducer is collecting echoes from gates 3 and 4, there is nothing at that x-y position, which remains a dark pixel.

There are about a dozen other AMI tool imaging modes that can be used to image MLCCs. One imaging mode non-destructively cross-sections through as many vertical planes in an MLCC as desired. Another imaging mode shows internal features in three dimensions. Another mode images the flatness of the MLCC's surface, and yet another mode images its internal features at single ultrasonic frequencies.

There is also a method, especially useful for high-power MLCCs, that

images all internal features capable of blocking ultrasound as acoustic shadows, as shown in Fig. 3. The detailed operation is shown in Fig. 4.

This method collects an echo only when a pulse has traveled to the bottom of the capacitor and back without encountering an imageable feature—as seen in the centre of the figure. The echo will be attenuated somewhat but will arrive in the gate assigned to it at precisely the right time. The acoustic image will show a grayish pixel at this x-y location.

At left in Fig. 4 is the path of a pulse that encountered a rounded void at a shallow depth. Ultrasound that was not scattered by the sloping sides of the void will be reflected in the direction of the transducer. But since the distance is less, it arrives far too early and is not collected. But no ultrasound will reach the bottom directly beneath the void. Areas around it will be reflected from the bottom and create a grayish pixel. In the acoustic image, then, the oval pulse will be black (no echoes received at the right time) but surrounded by grayish pixels.

The flat delamination at lower right has essentially the same fate. When a pulse strikes it, the ultrasound is reflected, but it arrives too soon to be in the proper gate. Like the void, it becomes a black acoustic shadow in a gray field.

The whole acoustic image will therefore show voids and delaminations as dark features—acoustic shadows. Since voids and delaminations are harmful at any depth, any capacitor exhibiting dark features will be discarded. Since the mere presence of defects is enough to disqualify such capacitors, the defect's depth is unimportant. **EFY**



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